

LM2 DDM

分清 PV, TV.
TV 不折现.

GGM

$$V_0 = \frac{D_1}{r-g}$$

不要管 D_0, D_1 关系.
只要后期 g 固定, 直接用 D_1 .

☆ 区分 payout ratio $(1-b) = \frac{\text{Div Paid}}{\text{NI}}$ 不是 Equity!
retention rate.

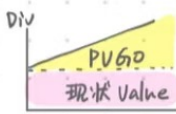
① $g = ROE \times b$ $g = \text{GDP} \% \text{ (长期)}$

$$g = \frac{\text{Net income} - \text{Dividends}}{\text{Net income}} \times \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total assets}} \times \frac{\text{Total asset}}{\text{Shareholders' equity}}$$

ROE

② $g > 0, = 0, < 0$. (优先股 $g=0, V_0 = \frac{D}{r}$) D 为 0 or 1, $\therefore g=0$ 都一样

③ 由公式计算 implied r or implied g . $r = \frac{D_1}{P_0} + g, \quad g = r - \frac{D_1}{P_0}$

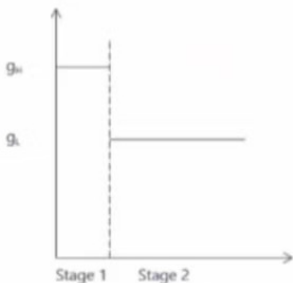
④ PVGO  $V_0 = \frac{E_1}{r} + \text{PVGO}$
 $\text{PVGO} = \frac{D_1}{r-g} - \frac{E_1}{r}$ 全分红假设

⑤ $\left\{ \begin{array}{l} \text{Leading P/E} = P_0/E_1 = \frac{1-b}{r-g} \\ \text{Trailing P/E} = P_0/E_0 = \frac{(1-b)(1+g)}{r-g} \end{array} \right.$

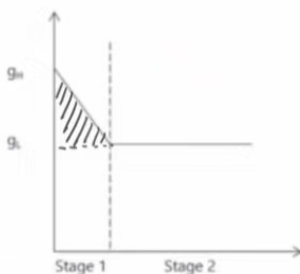
Fairly Value 时.
 r_e
Total Return = Div yield + Capital gain rate
要求回报率 = 总回报率

Multi-stage DDM

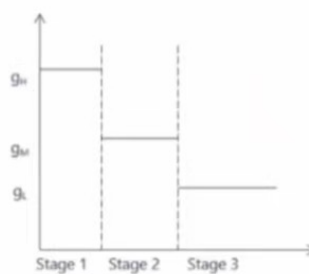
Two-stage DDM



H-model



Three stages model



计算 NPV 功能!
 $CF_0 = 0$
 $CF_1, \dots, \dots$
 $CF_n = D_n + TV_n$
CPT NPV: $I/Y = \text{rate}$

PV of Stage 1
Sum of PV of Div
+ PV of Stage 2
GGM

$$V_0 = \frac{D_0 \times (1+g_1) + [D_0 \times H(g_1 - g_2)]}{r - g_1}$$

永续 补三角
 $\uparrow$ $r-g_1$ $\uparrow$ $r-g_2$

PV of Stage 1
+ PV of Stage 2
+ PV of Stage 3

Dupont

$$ROE = \frac{\text{NI}}{\text{equity}} = \frac{\text{NI}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{sales}} \times \frac{\text{sales}}{\text{assets}} \times \frac{\text{assets}}{\text{equity}}$$

LM 3 Free Cash Flow Valuation

FCF

① 3优势 vs. Div: 有公司不发 Div / Align with profitability / "Control" perspective

② 选择: FCFE < 0 或 Capital Structure 有重大变动, 用 FCFE

③ 计算:

$$\begin{aligned} \text{FCFF} &= \text{EBIT} \times (1-T) + \text{NCC} - \text{FCInv} - \text{WCInv} \\ &= \text{NI} + \text{Int}(1-T) + \text{NCC} - \text{FCInv} - \text{WCInv} \\ &= (\text{EBITDA} - \text{D\&A}) \times (1-T) + \text{NCC} - \text{FCInv} - \text{WCInv} \\ &= \text{CFO} + \text{Int}(1-T) - \text{FCInv} \end{aligned}$$

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WCInv}$$

$$\text{FCFE} = \text{FCFF} - \text{Int}(1-T) + \text{Net Borrowing}$$

④ Items 计算

① $\text{NCC} = \text{D\&A}$

② $\text{WCInv} = \Delta \text{WC}$

$$\text{WC} = \text{CA} - \text{C.L}$$

Current ST Debt

CA: account receivable
inventory

CL: account payable
accrued tax & expense

不包含 notes payable!

③ $\text{FCInv} = \text{Capex} - \text{Proceeds}$

2种情形: No Disposal: $\text{Capex} = \text{Gross PPE}_{\text{末}} - \text{Gross PPE}_{\text{初}}$

未处置, proceeds = 0

Disposal: $\text{Capex} - \text{Proceeds}$

$$\text{Capex} = \text{Gross PPE}_{\text{末}} - \text{Gross PPE}_{\text{初}} + \text{Gross PPE}_{\text{disposal}}$$

$$\text{G/L} = \text{Proceeds} - \text{Book PPE}_{\text{disposal}}$$

④ Net Borrowing

2种方法

Debt 末 - Debt 初 (LT+ST)

Debt 包含 LT Debt, notes payable

$$(\text{WCInv} + \text{FCInv} - \text{Dep}) \times \text{Debt Ratio}$$

⑤ Preferred Stock (NI的选择)

$$\text{NI total} = \text{Div pre} + \text{NI common}$$

$$\text{FCFF} = \text{NI common} + \text{Div pre}$$

$$\text{FCFE} = \text{NI total} - \text{Div pre}$$

Valuation (同DDM)

① Single Stage Model

For FCFF valuation: $V_0 = \frac{\text{FCFF}_1}{\text{WACC}-g}$

Firm Value 只是 value of operating asset

注意: FCFE 算出来是 operating asset value 与 Firm Value.

Equity Value

② Two Stage Model

For FCFE valuation: $V_0 = \frac{\text{FCFE}_1}{r-g}$

③ Three Stage Model

LM 4. Multiples 定性为主

P/E

① 注意事项

Adjusted EPS

Exclude nonrecurring items

Normalize EPS

(剔除经济周期)

保持会计可比性

P/E → E/P

Portfolio P/E. 用 harmonic average 加权调和平均数

Average EPS

Average ROE

保留 size 等影响, better than average EPS

① 取倒数

② 取平均

③ 取倒数

② Justified P/E

$$P_0/E_0 = \frac{(1-b)(1+g)}{r-g}$$

< $\frac{b \cdot g}{r}$ 正相关
负相关

③ PEG

$$PEG = \frac{P/E}{g} \quad \text{越低越好}$$

算BV时:

Common shareholders' Equity 包含 Common Shares
以及 P/E!
不包含 Preferred Shares

P/B

Justified P/B

V_0/B_0

$$P_0/B_0 = 1 + \frac{ROE-r}{r-g} = \frac{ROE-g}{r-g}$$

Book Value per share $\times$ ROE = EPS

P/S

Justified P/S

V_0/S_0

$$P_0/S_0 = \frac{E_0}{S_0} \times \frac{P_0}{E_0} = \text{net margin} \times \frac{(1-b)(1+g)}{r-g} \quad \text{trailing}$$

$$\times \frac{1-b}{r-g} \quad \text{leading}$$

P/CF

没考过

P/Div or Div/P (Dividend Yield)

Justified Div Yield

D_0/V_0

$$D_0/P_0 = \frac{r-g}{1+g}$$

EV/EBITDA

① EV 计算:

$$EV = \text{MV of Common Stock} + \text{MV of Preferred Equity} + \text{MV of Debt} - \text{Cash and ST Investments}$$

② EBITDA 优缺点:

较NI更反映真实盈利能力, usually positive

$$\text{Invested Capital} = \text{MV of Common Stock} + \text{MV of Preferred Equity} + \text{MV of Debt}$$

$$\text{Unlevered Earnings (UE)} = EPS_t - E(CEPS_t)$$

$$\text{Standardized UE (SUE)} = \frac{EPS_t - E(CEPS_t)}{E[EPS_t - E(CEPS_t)]}$$

LM5 RI Valuation

Concept

股东: $\text{Residual Income} = NI - \text{Equity } BV_{\text{初}} \times r_e$
 All: $\text{Economic Value Added (EVA)} = \text{NOPAT} - \text{Capital}_{\text{初}} \times \text{WACC}$

公式

DCF
 $V_0 = \sum_{n=1}^K \frac{RI_n}{(1+r)^n} + B_0$

$MVA = MV - BV$ 股+债

RI 预测

g 不适用

$RI_1 = NI_1 - B_0 \times r_e$
 $= B_0 \times ROE - B_0 \times r_e$
 $= B_0 \times (ROE - r_e)$
 $RI_2 = B_1 \times (ROE - r_e)$
 $B_0 \rightarrow B_1 \rightarrow B_2 \rightarrow B_n \dots$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $RI_1 \quad RI_2 \quad RI_3 \quad RI_{n+1}$

$B_0 \xrightarrow{?} B_1$ RI Valuation 的前提.
 Clean Surplus Relation: 长期看, 影响 equity 的只有 R/E
 $\therefore B_0 + \Delta R/E = B_1$
 $B_0 + NI_1 - \text{Div}_1 = B_1$
 $B_0 + B_0 \times ROE - B_0 \times ROE \times \text{Payout Ratio} = B_1$

Valuation

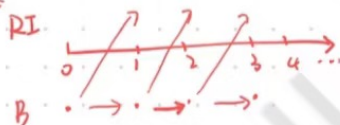
① Single-stage
 ① $V_0 = \frac{RI_1}{r-g} + B_0 = \frac{B_0 \times (ROE-r)}{r-g} + B_0$
 ② V_0 已知, 算 implied r.g.
 ③ 算 Justified P/B. $\frac{\frac{B_0 \times (ROE-r)}{r-g} + B_0}{B_0} = \frac{ROE-r}{r-g} + 1$

② Two-stage

$V_0 = B_0 + \text{PV of Stage 1 RI} + \text{PV of Stage 2 RI}$

$B_0 \rightarrow B_1 \rightarrow B_2 \rightarrow B_3$
 $\downarrow \quad \downarrow \quad \downarrow$
 $RI_1 \quad RI_2 \quad RI_3 \rightarrow \text{PV}$
 一期一期推, 折现加总.

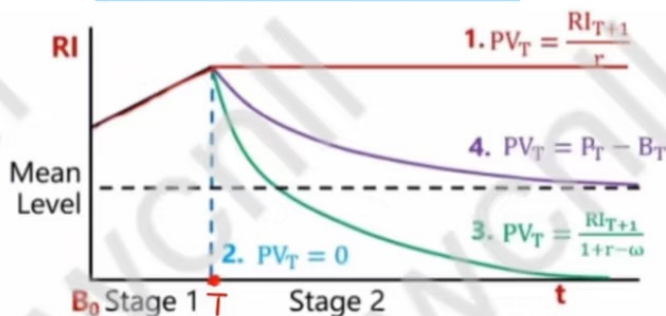
做题:



四种情况

① 永续 $PV_T = \frac{RI_{T+1}}{r}$ $g=0$
 ② $RI \rightarrow 0$ $PV_T = 0$
 ③ $RI \downarrow$ $PV_T = \frac{RI_{T+1}}{r-g} = \frac{RI_{T+1}}{1+r-\omega}$
 ④ $RI \downarrow$ 行业平均 $PV_T = P_T - B_T$

persistence factor $\omega = 1+g$



LM6 Private Company Valuation

F/S 调整

- 1 Normalized Earnings
 - 剔除 nonrecurring
 - 做高 Expense 扣除
 - 剔除 non-operating asset
- 2 Synergy
 - Strategic: + synergy
 - Financial (Non Strategic): 不+
- 3 Earnings $\rightarrow$ CF DCF 用 FCFF or FCFE.

Valuation

- 可类比上市公司 Valuation
- 1 Income Approach
 - GEM
 - $Earnings = E_{W.C.} + E_{F.A.} + E_{I.A.}$
 - $V_{I.A.} = \frac{E_{I.A.} \times (1+g)}{r-g}$ 更适用于小公司/关心 V.I.A./关心 Total Capital
 - $Firm Value = V_{I.A.} + W.C. + F.A.$
 - r 是 I.A. 的回报率
 - CCM 特指用于 Private Company 估值. 永续 $\frac{CF_1}{Cap Rate} = \frac{CF_1}{r-g}$
 - 2-stage
 - 2 Market Approach
 - 重点在 benchmark
 - GPCM - Guideline Public Company: + Control Premium
 - GTM - Guideline Transaction: 无需 Control Premium
 - PTM - Prior Transaction: 前轮融资
 - 3 Asset-Based Approach - $FV \text{ of Asset} - PV \text{ of Liability}$, 计算 Liquidation Value
 - 适用于
 - Financial Company
 - Natural Resources Company
 - Intangible Asset 少
 - Liquidation

Discount

- 1 Discount for lack of control - $DLOC = 1 - \frac{1}{1 + \text{Control premium}}$
 - 2 Discount for lack of marketability Liquidity - $DLOM$ 不用算
- $Total Discount = 1 - [(1 - DLOC)(1 - DLOM)]$
- 使用中 $\times (1 - Total Discount)$.